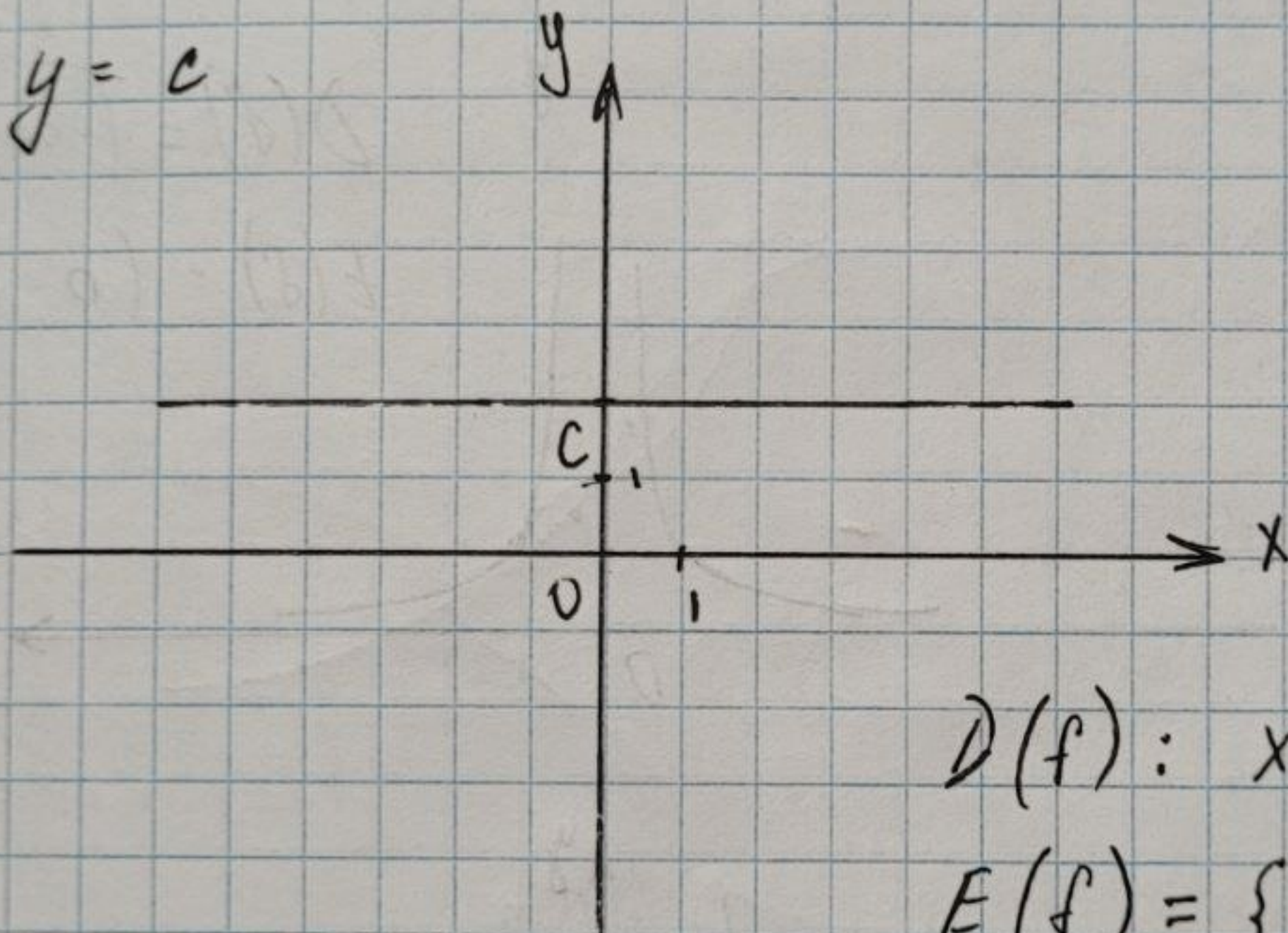


1) Постоянная ф-ция

$$y = c$$



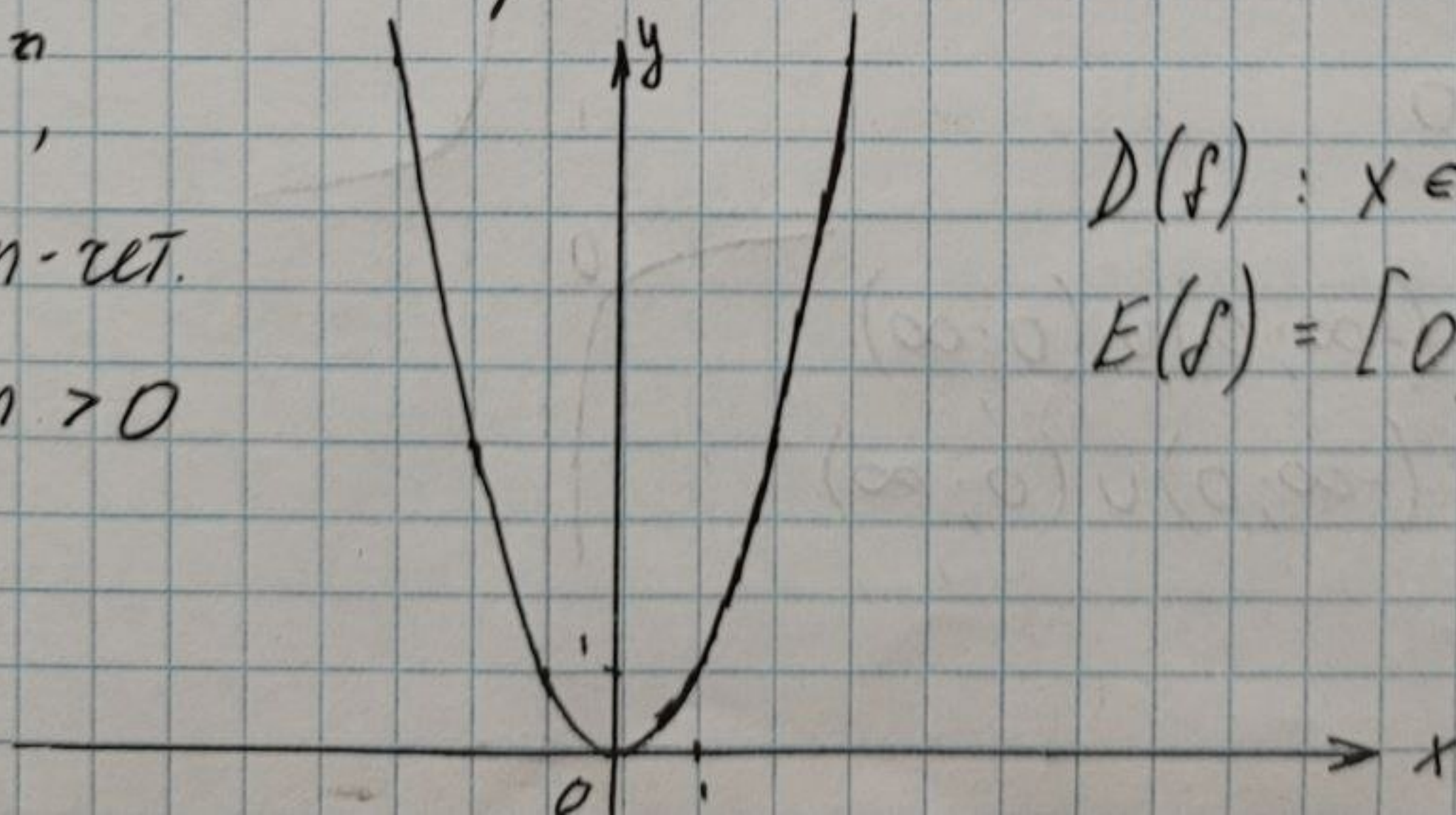
$$D(f): x \in \mathbb{R}$$

$$E(f) = \{c\}$$

2) Степенная ф-ция

a) $y = x^n$,
где n - чет.

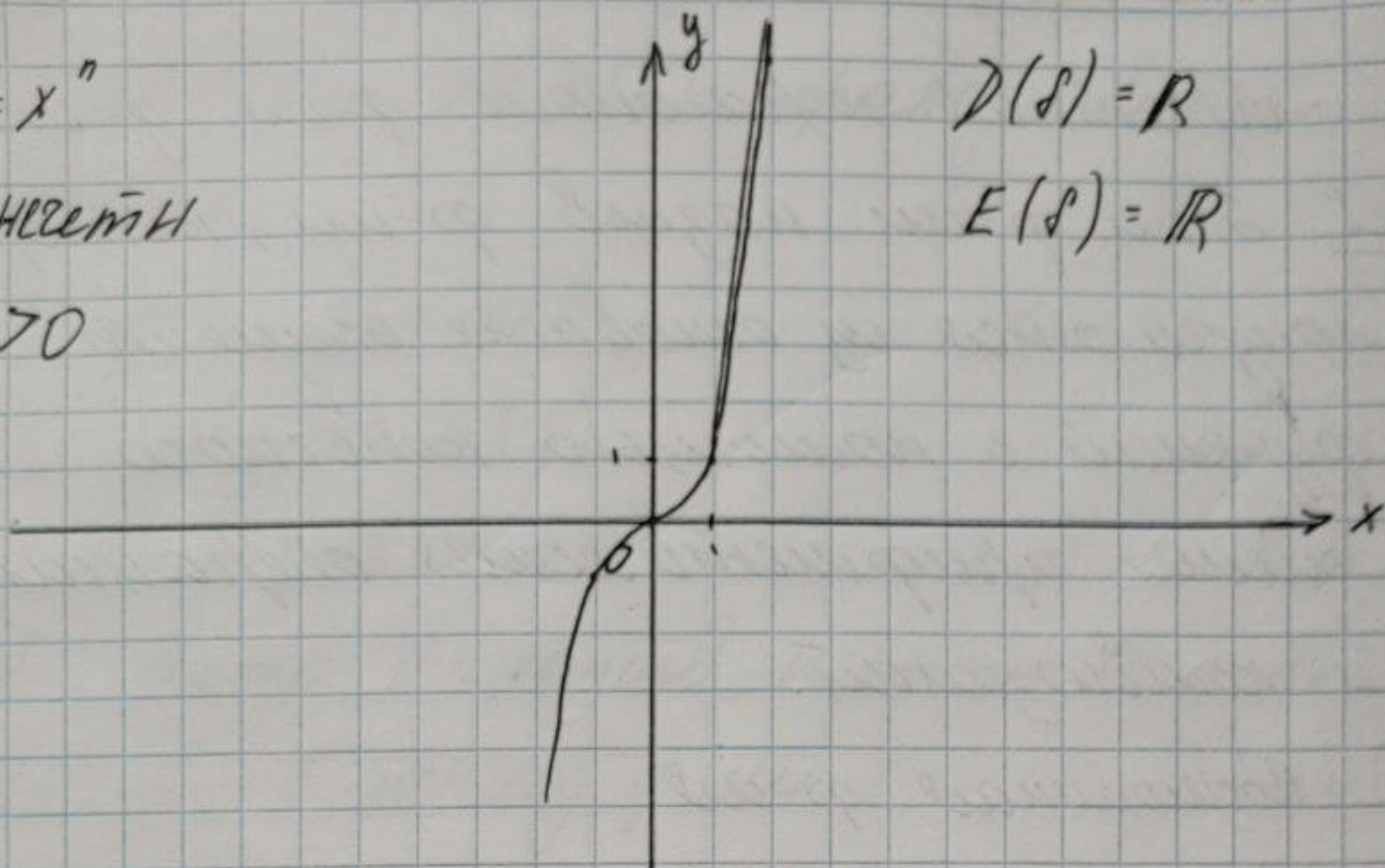
$$n > 0$$



$$D(f): x \in \mathbb{R}$$

$$E(f) = [0; \infty)$$

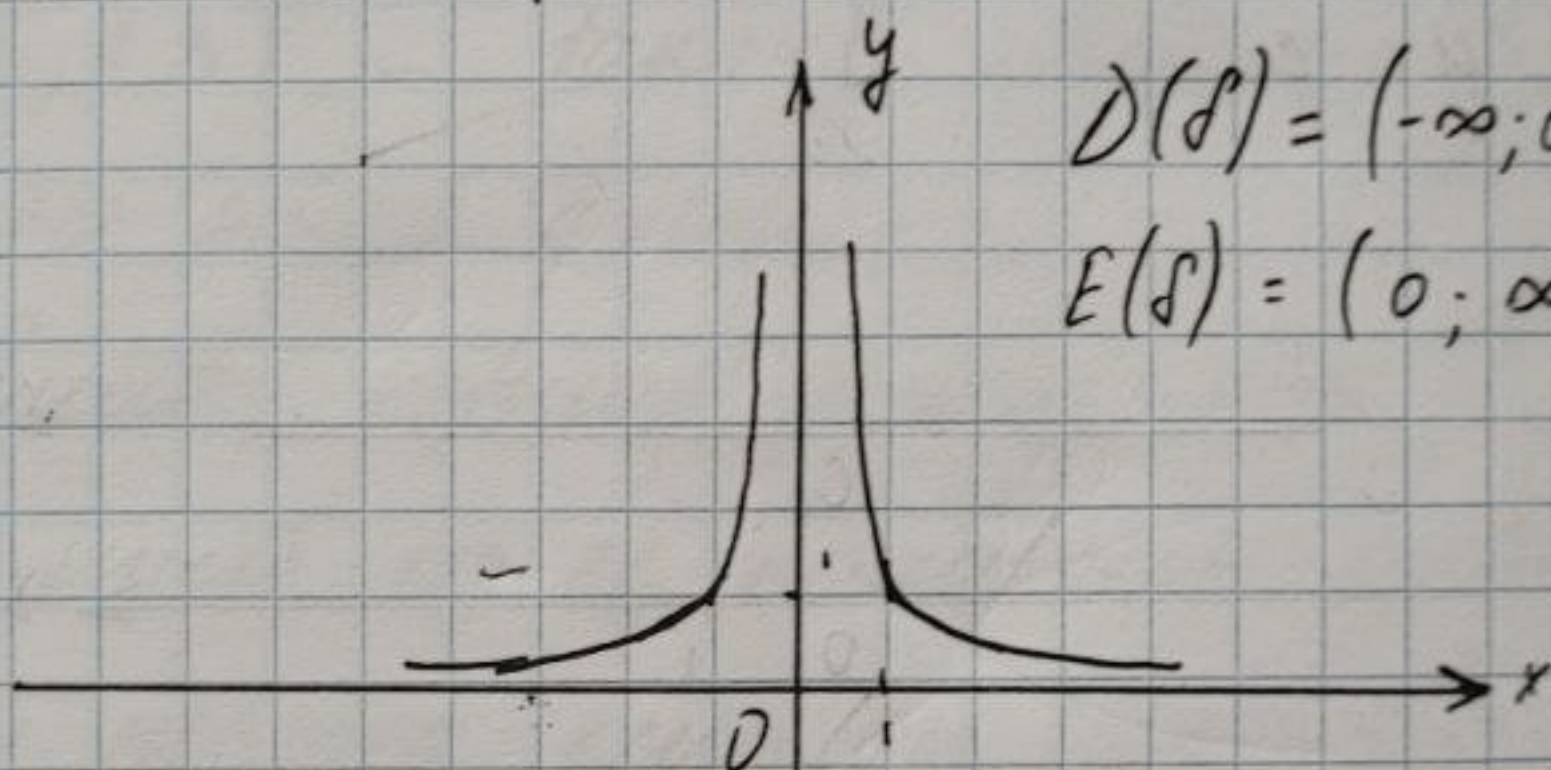
3) 6) $y = x^n$
 n - нечетн
 $n > 0$



$$D(f) = \mathbb{R}$$

$$E(f) = \mathbb{R}$$

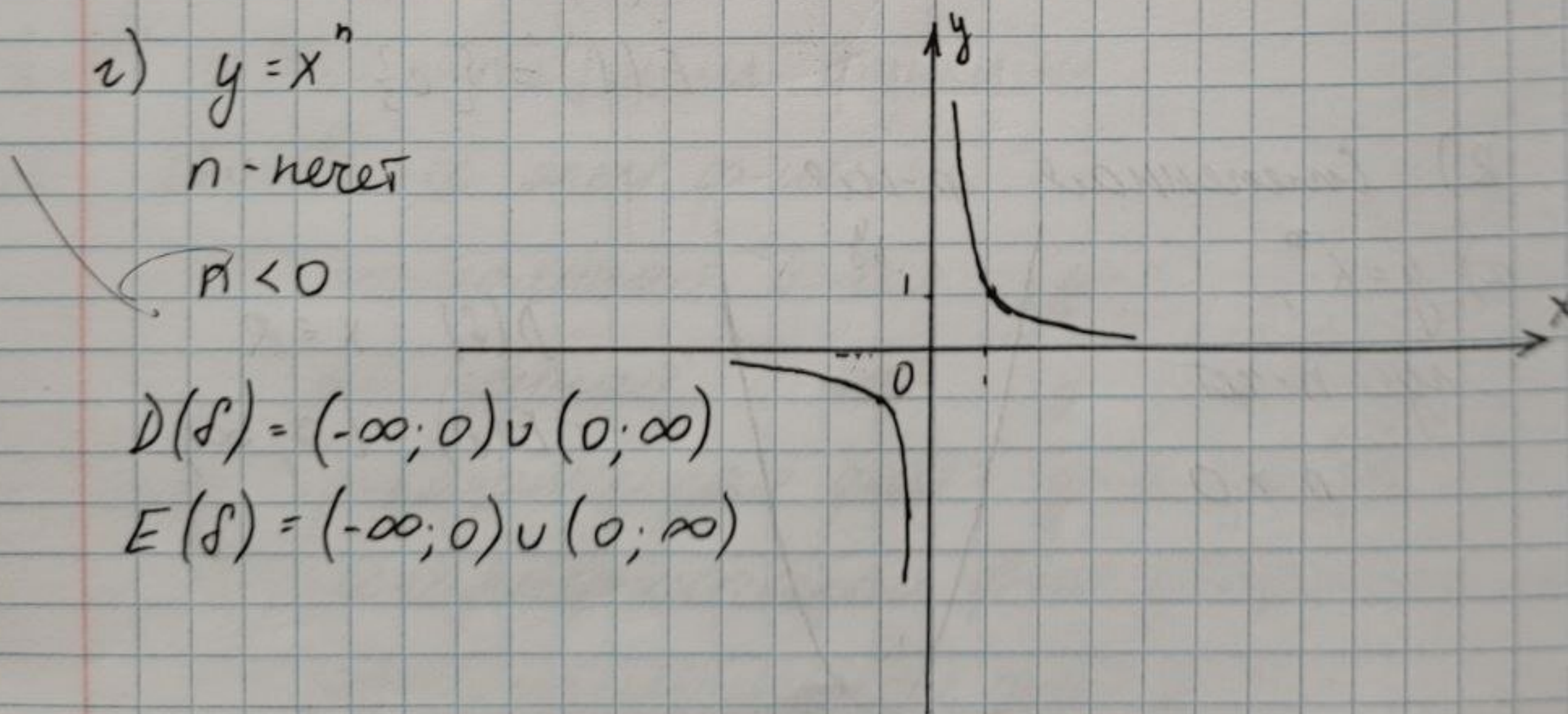
8) $y = x^n$
 n - четн
 $n < 0$



$$D(f) = (-\infty; 0) \cup (0; \infty)$$

$$E(f) = (0; \infty)$$

2) $y = x^n$
 n - четн
 $n < 0$



$$D(f) = (-\infty; 0) \cup (0; \infty)$$

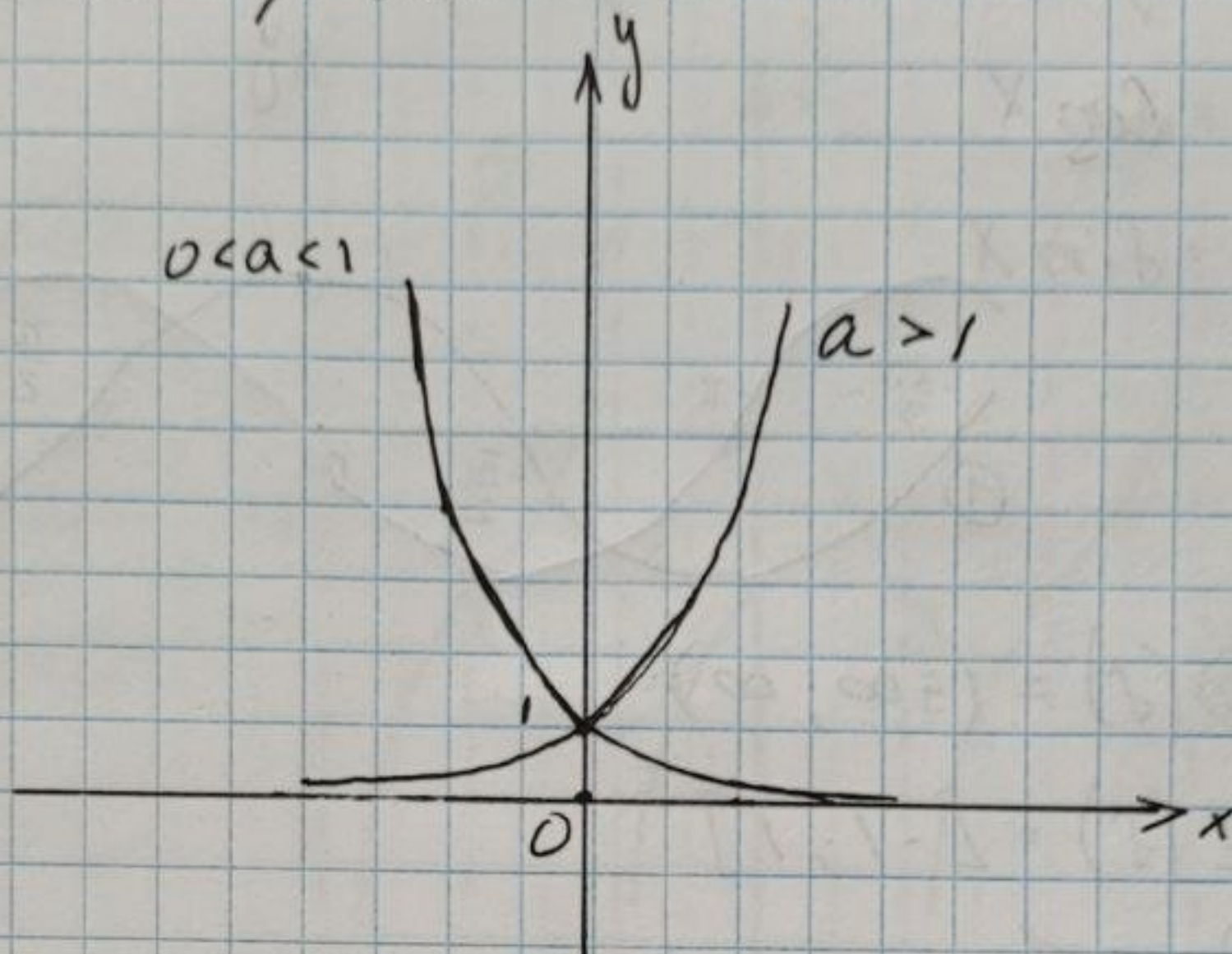
$$E(f) = (-\infty; 0) \cup (0; \infty)$$

3) Показательная ф-ция

$$y = a^x$$

$$D(f) = \mathbb{R}$$

$$E(f) = (0; \infty)$$



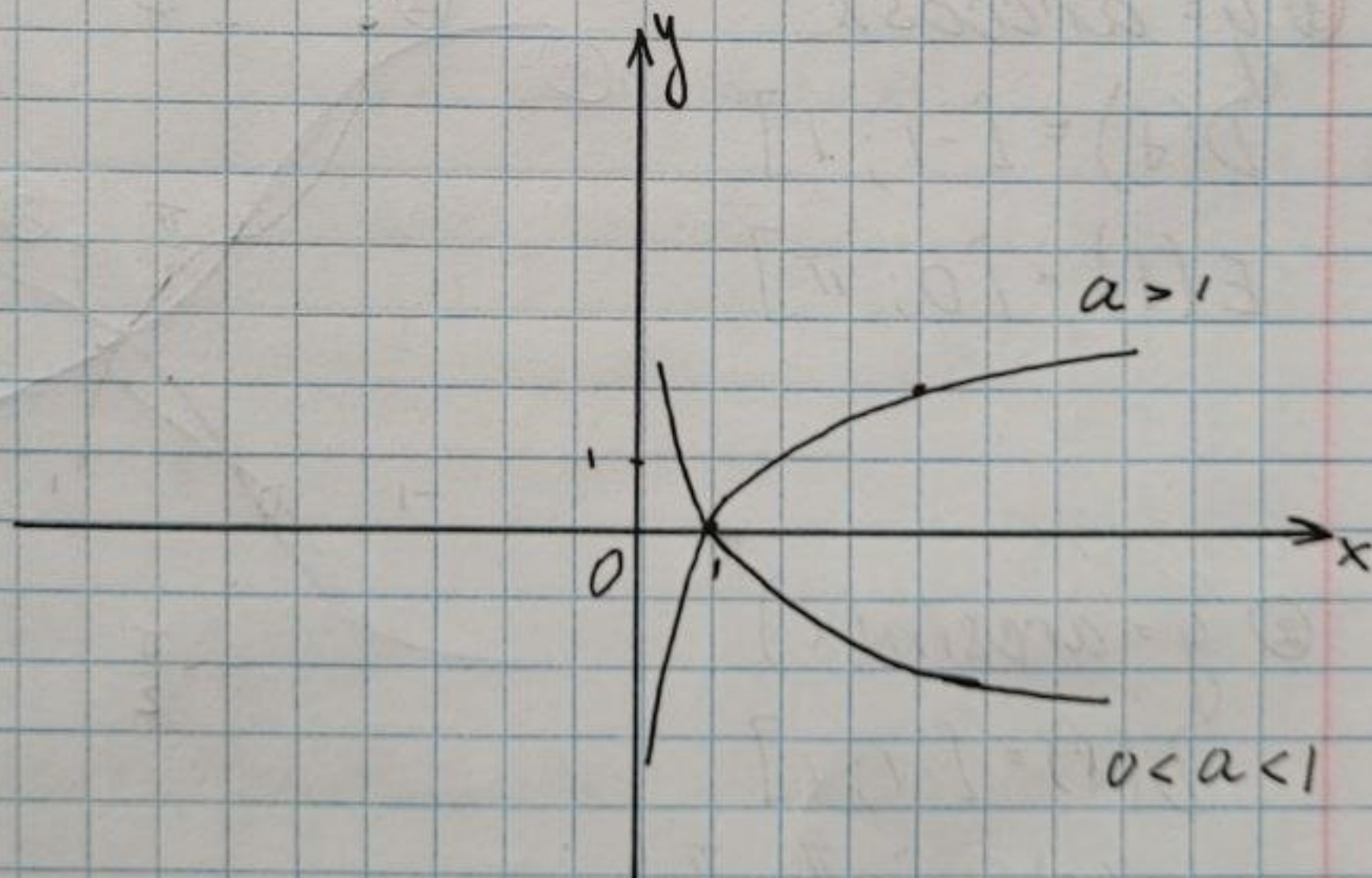
4) Логарифмическая ф-ция

$$y = \log_a x$$

$$a > 0 \quad a \neq 1$$

$$D(f): x > 0$$

$$E(f): y \in \mathbb{R}$$



5) Тригонометрические функции

① $y = \operatorname{tg} x$

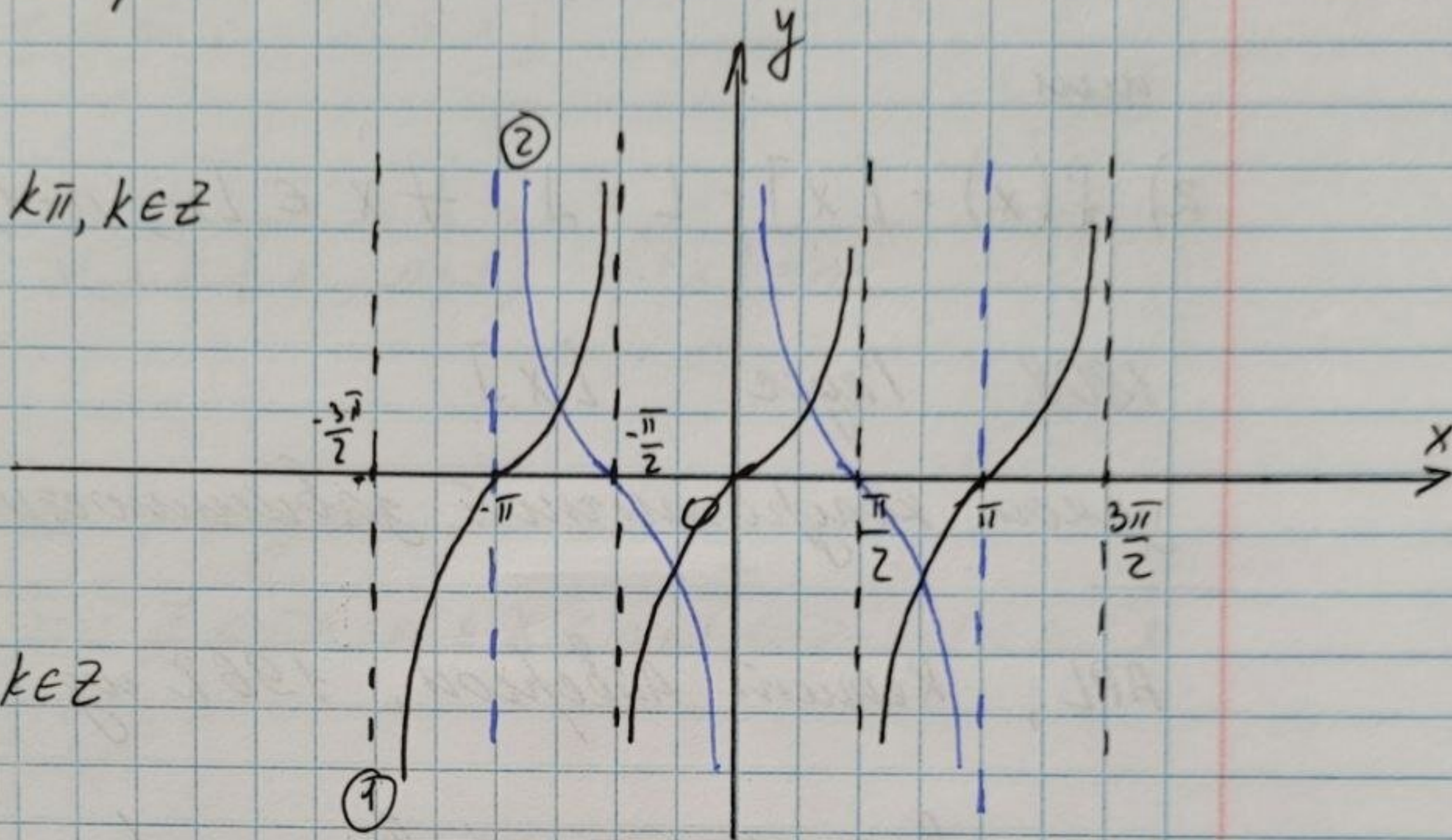
$$D(f) = x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$$

$$E(f) : y \in \mathbb{R}$$

② $y = \operatorname{ctg} x$

$$D(f) : x \in k\pi, k \in \mathbb{Z}$$

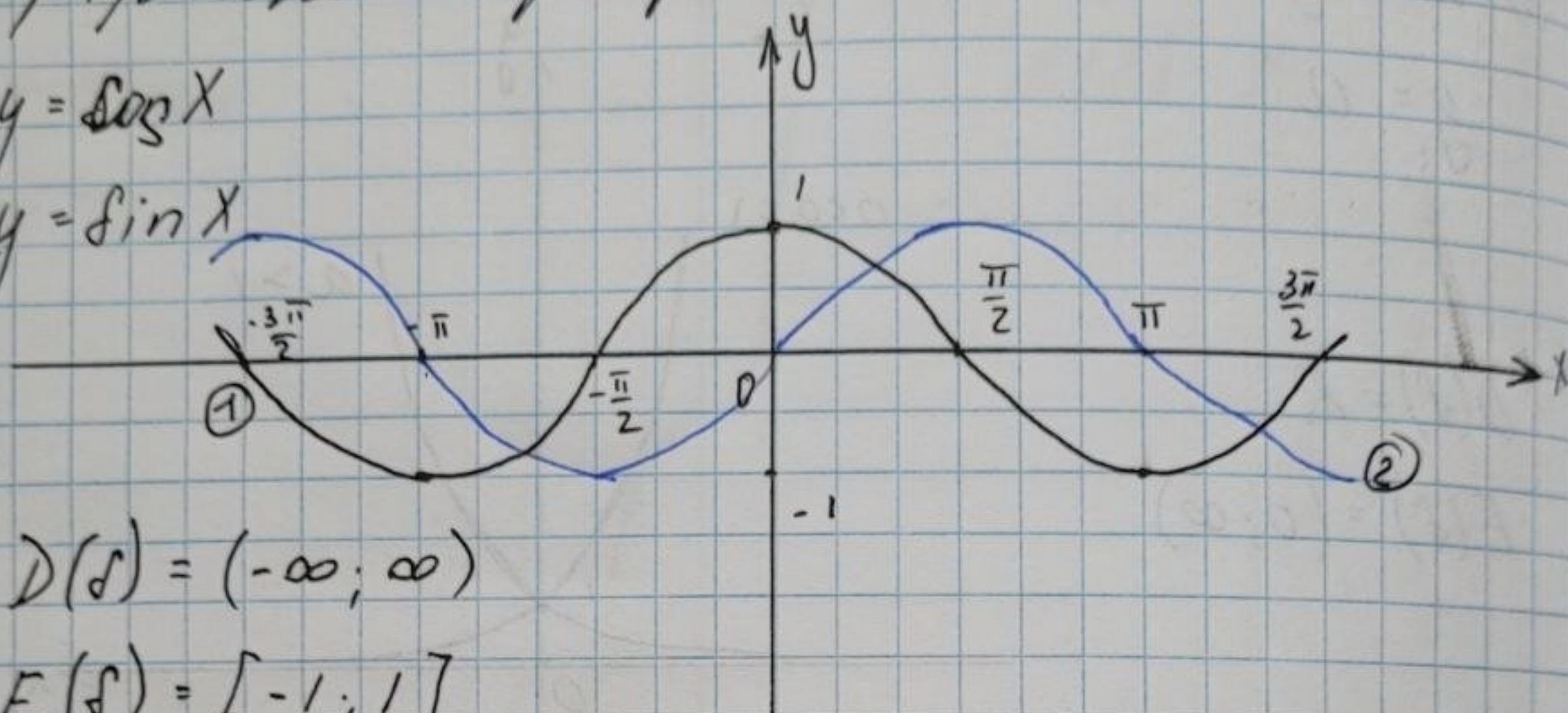
$$E(f) : y \in \mathbb{R}$$



В 5) Тригонометр. ф-ии

① $y = \cos x$

② $y = \sin x$



$D(f) = (-\infty; \infty)$

$E(f) = [-1; 1]$

6) Обратн. тригон. ф-ии

① $y = \arccos x$

$D(f) = [-1; 1]$

$E(f) = [0; \pi]$

② $y = \arcsin(x)$

$D(f) = [-1; 1]$

$E(f) = [-\frac{\pi}{2}; \frac{\pi}{2}]$

③ $y = \operatorname{arctg}(x)$

$D(f) = (-\infty; \infty)$

$E(f) = (-\frac{\pi}{2}; \frac{\pi}{2})$

④ $y = \operatorname{arccotg}(x)$

$D(f) = (-\infty; \infty)$

$E(f) = (0; \pi)$

